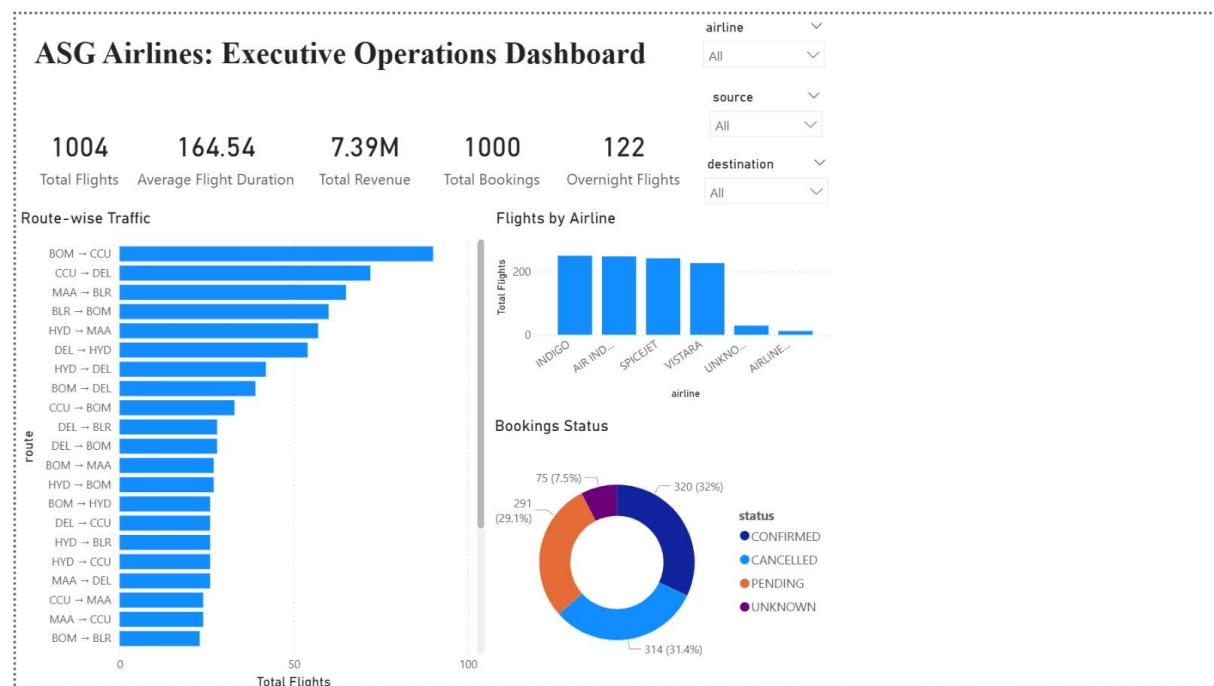


ASG Airlines – Power BI Dashboard Report

Submitted By: Subiksha N

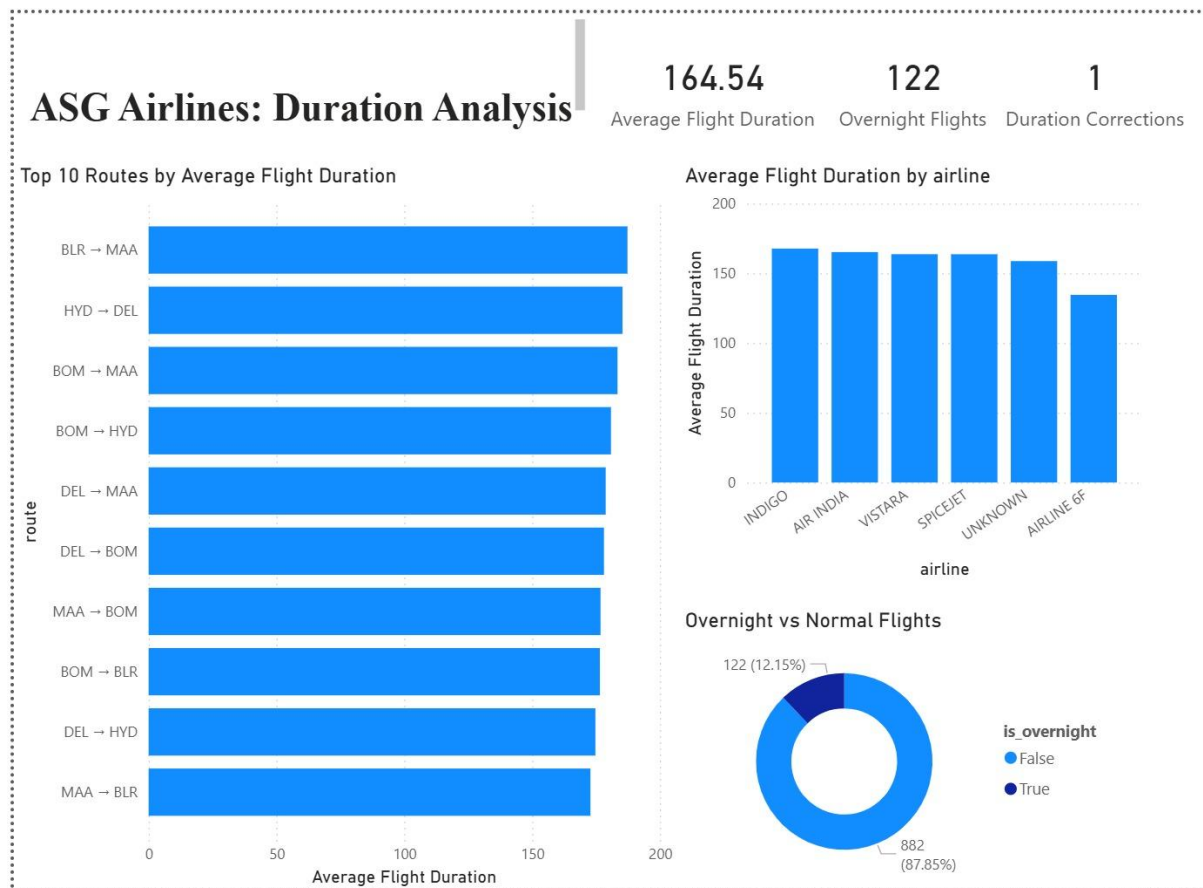
Page 1 Executive Operations Dashboard

- This page provides the executive-level summary of airline operations. KPI cards provide an immediate view of the overall scale and performance of the dataset.
- KPI cards: Total Flights, Average Flight Duration, Total Revenue and Overnight Flights.
- Flights by Airline chart compares the operational flight volume across airlines.
- Route Traffic visual highlights traffic distribution across source-destination routes.
- Booking Status Distribution shows confirmed, cancelled, pending and unknown booking statuses.
- Slicers allow users to filter the report by Airline, Source and Destination.
- Operational use: management can quickly understand flight volume, booking activity, route demand, airline distribution and overnight operations.



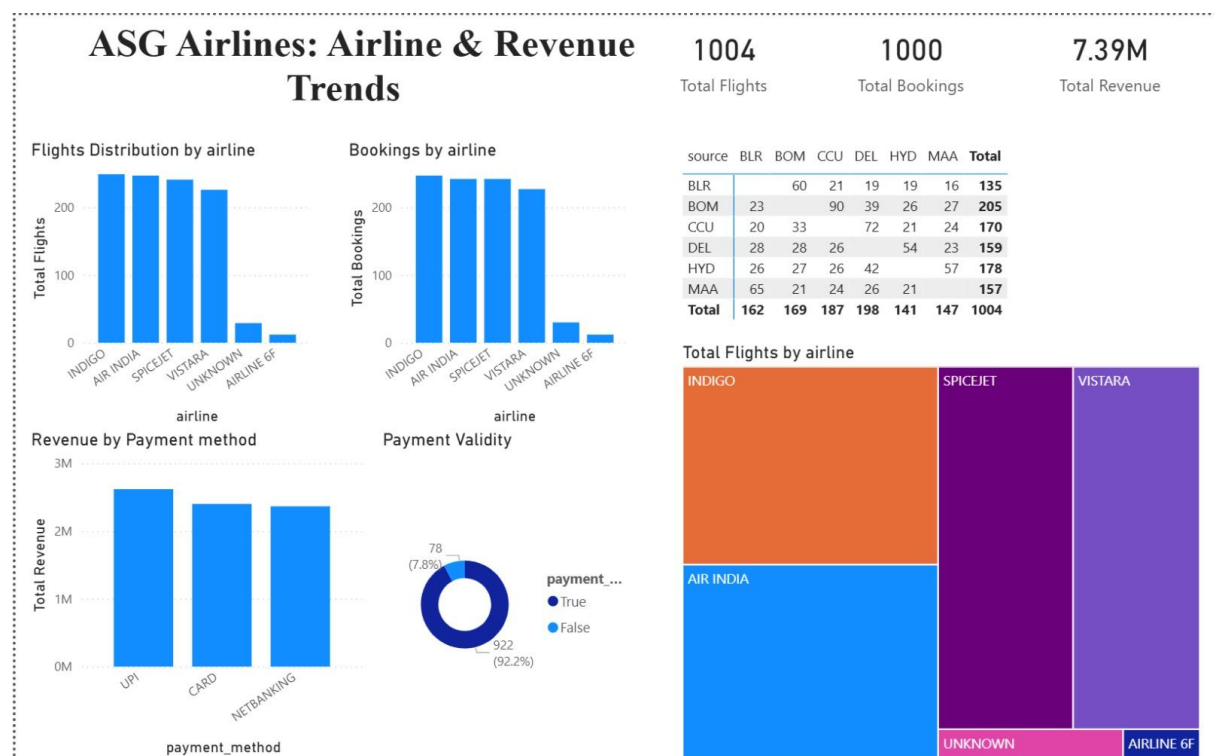
Page 2 Duration Analysis

- This page focuses on flight duration patterns and overnight operations. It helps identify airlines and routes associated with longer average flight times.
- Average Flight Duration KPI.
- Overnight Flights KPI.
- Duration Corrections KPI.
- Average Duration by Airline comparison.
- Top 10 Routes by Average Duration.
- Overnight versus Non-Overnight flight distribution.
- Operational use: the page supports route planning and operational analysis by showing duration differences between airlines and routes and highlighting overnight operations.



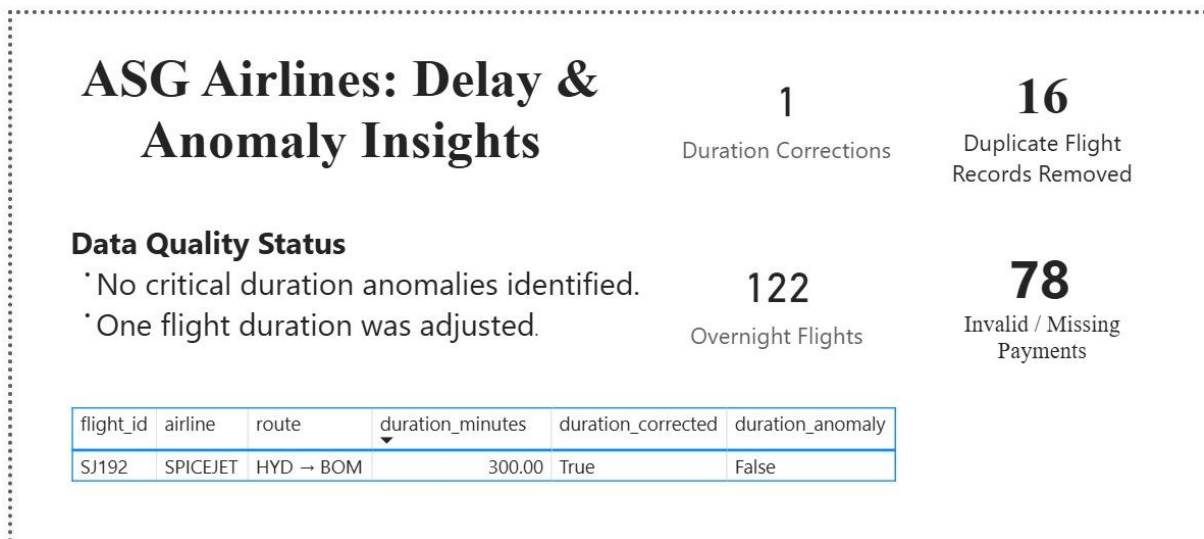
Page 3 Airline & Revenue Trends

- This page combines operational activity with booking and payment information to provide an airline and revenue-oriented view.
- Total Flights, Total Bookings and Total Revenue KPIs.
- Flights by Airline.
- Bookings by Airline.
- Revenue by Payment Method.
- Payment Validation Status.
- Source-to-Destination matrix.
- Airline comparison using a treemap.
- Operational use: decision-makers can compare airline activity, booking volumes and payment/revenue patterns while identifying the routes contributing to operations.



Page 4 Delay & Anomaly Insights

- This page provides operational quality and anomaly indicators. The source dataset does not contain a dedicated delay-minutes field, so the report does not create or estimate a synthetic delay metric.
- Overnight Flights: 122.
- Duration Corrections: 1.
- Corrected flight identified by the pipeline: SJ192.
- Duplicate records removed: 16.
- Invalid or missing payments: 78.
- Data-quality status indicators.
- Operational use: this page allows users to identify data-quality issues and operational exceptions that may require investigation.



Interactivity and User Experience

- Slicers allow filtering by airline, source and destination.
- Power BI cross-filtering enables users to select a category in one visual and examine its effect on other visuals.
- KPI cards provide high-level metrics before users drill into detailed charts.
- Charts and tables are grouped by business purpose so operational, duration, revenue and quality analysis are separated.
- The four-page structure keeps the report readable while supporting progressively deeper analysis.

Visualization Strategy

Visualization	Purpose	Business question
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KPI Cards	Headline metrics	What is the current operational scale?
Bar Charts	Category comparison	Which airline/route has the highest activity or duration?
Donut Charts	Part-to-whole analysis	How are bookings or overnight flights distributed?
Matrix	Detailed route comparison	How does activity vary between source and destination?
Treemap	Relative category size	Which airlines contribute the largest operational volume?
Slicers	Interactive filtering	How do results change for a selected airline or route?

Business Insights Supported by the Report

- The cleaned dataset contains 1,004 operational flights across 30 routes and 6 airlines.
- Average flight duration is 164.54 minutes.
- 122 flights are identified as overnight operations.
- The pipeline recorded 1 duration correction and no remaining extreme-duration anomalies.
- There are 1,000 bookings, enabling comparison of operational flights with booking activity.
- Valid payment records contribute 7,385,142.98 to the Total Revenue KPI.
- 78 payment records are flagged as invalid or missing and can be investigated through the quality view.
- The dashboard does not fabricate delay information because the source does not provide delay minutes.

Dashboard Deliverables

- Power BI report: ASGAirlinesdashboard find.pbix
- Dashboard Page 1 screenshot: dashboard_page1.jpg
- Dashboard Page 2 screenshot: dashboard_page2.jpg
- Dashboard Page 3 screenshot: dashboard_page3.jpg
- Dashboard Page 4 screenshot: dashboard_page4.jpg

- The interactive .pbix file is the primary dashboard deliverable; the four screenshots provide static evidence of the completed report pages.

Conclusion

The Power BI report converts the Gold analytical datasets into an interactive operational reporting solution. It combines executive KPIs, duration analysis, airline and revenue trends, and data-quality/anomaly insights in a four-page structure suitable for operational decision-making.